

High-Performance Mixed-Dimensional Reconfigurable Field-Effect Transistors with Enhanced Polarity Control

A. Echresh¹ S. Ghosh¹ M. Khan¹ P. Chava¹ K. Watanabe³ T. Taniguchi³
S. Prucnal¹ R. Hübner¹ Y. Georgiev¹ A. Erbe^{1,2}

¹Institute of Ion Beam Physics and Materials Research, Helmholtz-Zentrum Dresden-Rossendorf (HZDR), 01328 Dresden, Germany

²Technische Universität Dresden, Centre for Advancing Electronics (CfAED), 01069 Dresden, Germany

³National Institute for Materials Science, 1-1 Namiki, Tsukuba 305-0044, Japan

The continued physical downscaling of CMOS technology is approaching fundamental limits, motivating the exploration of alternative device concepts and material platforms. One promising approach is the reconfigurable field-effect transistor (RFET), which can be dynamically programmed to exhibit n- or p-type behavior through electrostatic control [1]. In this work, we present a novel mixed-dimensional RFET concept based exclusively on one-dimensional (1D) silicon (Si) nanowires integrated with two-dimensional (2D) materials [2]. The device architecture consists of intrinsic Si nanowires contacted with nickel (Ni), followed by annealing to form silicided regions. The resulting silicide–Si interfaces create Schottky junctions, whose carrier injection properties are modulated by appropriate gate architectures. Ambipolar transport is achieved via electrostatic control using back-gate or top-gate configurations. The primary objective of this study is the optimization of Si nanowire RFETs employing hexagonal boron nitride (hBN) as a 2D dielectric and encapsulation layer instead of conventional thermally grown oxide. Owing to its atomically smooth surface, absence of dangling bonds, chemical stability, and dielectric constant comparable to SiO₂, hBN represents a suitable gate dielectric for nanoscale devices. Top-down fabrication of the Si nanowires is carried out using electron beam lithography and reactive ion etching [3], while controlled formation of silicided Schottky junctions is achieved by flash-lamp annealing [4]. Mechanically exfoliated hBN flakes (5–10 nm), characterized by atomic force microscopy and transmission electron microscopy, are transferred onto the devices using a dry viscoelastic stamping technique. Cross-sectional energy-dispersive X-ray spectroscopy confirms the element distribution. Electrical characterization demonstrates enhanced ambipolar and unipolar transfer characteristics and a significant improvement in subthreshold swing for hBN-encapsulated Si RFETs, particularly under top-gate operation, highlighting the effectiveness of 2D dielectric integration in mixed-dimensional silicon-based reconfigurable devices [5].

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